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15 October 1964
SUPERSEDING
FA-PD-M1-2298, Rev. 1
14 June 1962

MILITARY SPECIFICATION

FUZE, MECHANICAL TIME, M542, ASSEMBLY WITH IGNITER

1. SCOPE

1.1 This specification covers one type of mechanical time functioning fuze for use with large caliber artillery ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

- MIL-A-2550 - Ammunition and Special Weapons, General Specification for
- MIL-C-10464 - Cans, Hermetic Sealing, Metal Light Gage Tear-strip Type
- MIL-P-14100 - Primers, Stab Type, Loading, Assembling and Packing

STANDARDS

Military

- MIL-STD-8 - Dimensioning and Tolerancing
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-350 - Jolt Test for Use in Production of Fuzes
- MIL-STD-351 - Jumble Test for Use in Production of Fuzes
- MIL-STD-352 - Forty (40) Foot Drop Test for Use in Production of Fuzes
- MIL-STD-353 - Transportation-Vibration Test for Use in Production of Fuzes
- MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
- MIL-STD-629 - Surface Roughness Values Preferred for Army Ordnance Material

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Munitions Command
X-PA-PD-999 - Igniter Assembly for Shell, 8 Inch, T317E1 Loading, Assembly and Packing

X-PA-PD-998 - Igniter Tube Assembly for Shell, 8 Inch, T317E1 - Metal Parts

DRAWINGS

Munitions Command
F8595229 - Fuze Mechanical Time, MT, M542, Assembly with Igniter
F8794405 - Box, Packing, Ammunition for Fuze
F8794406 - Ammunition Container, Metal, for Fuze
F10523262 - Arrangement of Timers for Testing M542 Fuze
IEL-8595229 - Inspection Equipment List for Fuze, MT, M542 Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications, except as provided in 6.1 and 6.1.1.

3.2 Initial production samples.- Unless otherwise directed by the contracting officer, an initial production sample shall be required (see 4.2).

3.3 Environmental requirements

3.3.1 Jolt.- The fuze shall comply with MIL-STD-350.

3.3.2 Jumble.- The fuze shall comply with MIL-STD-351, except that the fuze shall be jumbled in a metal packing container.

3.3.3 Forty-foot drop.- The fuze shall comply with MIL-STD-352, except that the fuze shall be dropped in a metal packing container.

3.3.4 Transportation-vibration.- The fuze shall comply with MIL-STD-353. The operability of the vibrated fuze shall be determined by ballistic firing under the following conditions:

Weapon	Charge	Fuze Setting	Temperature
8" Howitzer	Zone 7	55 Seconds	$70^{\circ} \pm 10^{\circ}$ Fahrenheit(F)

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3.4 Explosive train.- The explosive train for each timer in the assembled fuze shall be independently capable of supplying sufficient energy to detonate the igniter.

3.5 Dynamic regulation.- The movement subassembly shall have a timing accuracy of ± 0.07% from 10 to 100 seconds when regulated at the frequency specified on the applicable drawing and rotated at a spin speed of 6000 ± 25 revolutions per minute (rpm).

3.6 Ballistic requirements.- The fuze, modified for testing in accordance with Drawing F10523262, shall detonate the test round (Drawing D10523269) within the time limits specified in Table I, when the temperature of the fuze, projectile and propellant is maintained at 70° ± 10° (F), except as otherwise specified herein.

3.6.1 Low temperature.- The fuze, modified for testing in accordance with Drawing F10523262, shall detonate the test round (Drawing D10523269) when the temperature of the fuze, projectile and propellant is maintained at -40° -5° F and fired as follows:

Weapon	Charge	Fuze Setting
8" Howitzer	Zone 1	20 seconds

3.6.2 High temperature.- The fuze, modified for testing in accordance with Drawing F10523262, shall detonate the test round (Drawing D10523269) when the temperature of the fuze, projectile and propellant is maintained at 160° +5° F and fired as follows:

Weapon	Charge	Fuze Setting
8" Howitzer	Zone 7	55 seconds

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.7.1.1 The firing arm assembly shall contain no fracture of the upright bend.

3.7.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and the provisions of MIL-A-2550.

3.7.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.7.4 Foreign matter.- No part of assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The cleaning methods and agents shall not adversely affect the finish nor contaminate the parts. The contractor shall maintain an environment conducive to cleanliness of all parts and assemblies.

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Table I

Weapon	Charge Zone	Set Time (Sec)		Limits for Indiv. Rounds (Sec)	Limits for Sample Mean Time (Sec) Sample Size 30	29	28	27	26
8" Howitzer	1	20	UL	21.029	20.600	20.602	20.604	20.606	20.608
			LL	19.799	20.228	20.226	20.224	20.222	20.220
					33	32	31	30	29
8" Howitzer	7	55	UL	56.216	55.516	55.518	55.520	55.523	55.526
			LL	54.366	55.066	55.064	55.062	55.059	55.056

Weapon	Charge Zone	Set Time	Upper Limits for Sample Std. Dev. (Sec) Sample Size 30	29	28	27	26
8" Howitzer	1	20	.238	.239	.240	.241	.243
			33	32	31	30	29
8" Howitzer	7	55	.380	.381	.383	.385	.387

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 lxcept as specified herein.

4.2 Initial production sample.- Prior to the initiation of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected for all the requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted. The initial production sample shall consist of forty-five (45) fuzes plus listed component parts and assemblies, to be tested as follows:

4.2.1 Ballistic tests.- The Government will perform the following ballistic tests on 63 test fuzes (6.3.4) derived from 21 disassembled fuzes.

Weapon (Howitzer)	Charge (Zone)	Test Fuze Setting	Test Fuze Quantity	Requirement Paragraph	Rejection Criteria	Test Procedure
8"	1	20 Sec	30	3.6	4.3.3.14	4.4.19
8"	7	55 Sec	33	3.6	4.3.3.14	4.4.19

4.2.2 Environmental tests.- Fifteen (15) fuzes shall be subjected to the following environmental tests:

Environmental Test	Quantity	Requirement Paragraph	Acceptance Criteria	Test Procedure
Jolt	3	3.3.1	4.3.3.13	4.4.17.1
Jumble	4	3.3.2	"	4.4.17.2
Forty (40 Foot Drop)	5	3.3.3	"	4.4.17.3
Transportation-Vibration	3	3.3.4	"	4.4.17.4

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4.2.3 Environmental-ballistic tests.- The government will subject twenty-seven (27) test fuzes (6.3.4.) derived from nine (9) disassembled fuzes to the following environmental-ballistic tests.

Environmental Ballistic Test	Weapon (Howitzer)	Charge (Zone)	Test Fuze Setting	Test Fuze Quantity	Requirement Paragraph	Rejection Criteria	Test Procedure
Post-Vib.	8"	7	55 Sec.	9	3.3.4	4.3.3.13.1	4.4.18
Low-Temp.	8"	1	20 Sec.	9	3.6.1	4.3.3.13.1	4.4.18
High-Temp.	8"	7	55 Sec.	9	3.6.2	4.3.3.13.1	4.4.18

4.2.4 Hardness tests.- A sample of twenty each of the component parts listed in paragraph 4.3.3.1 shall be submitted for hardness tests according to the test procedure denoted in 4.4.1.

4.2.5 Static load and torque tests.- A sample of 25 each of the assemblies listed in 4.3.3.2 shall be submitted for static load and assembly torque tests according to the test procedures denoted in 4.4.2 and 4.4.3, respectively.

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105, or MIL-STD-414, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the criteria in Table II shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Defect Classification	Acceptable Quality Level (AQL)
Major	AQL = 0.25 percent defective
Minor	AQL = 0.40 percent defective

4.3.2.1 Firing Arm Assembly, Drawing C8593094, covering a detail of Drawing F8595229.

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Defect	Method of Inspection
CRITICAL	
1. Firing arm upright fractured, broken or deformed	Visual
2. Spinning or staking of firing arm to firing arm shaft, inadequate	Visual
MAJOR	
101. Insecure or improper assembly	Manual & Visual
102. Metal defective	Visual
MINOR	
None defined	

4.3.2.2 Rotor Assembly, Drawing 8594410, covering a detail of Drawing F8595229.

CRITICAL	
1. Primer above flush of rotor surface	Visual
MAJOR	
201. Insecure assembly	Manual
202. Primer installed incorrectly (red end facing cushion)	Visual
203. Staking missing or inadequate	Visual
204. Parts missing or improperly assembled	Visual
MINOR	
None defined	

4.3.2.3 Delay Arming Mechanism Assembly, Drawing F8594387, covering a detail of Drawing F8595229.

CRITICAL	
1. Rotor in armed position (aligned with firing hole)	Visual
2. Escapement lever pallet(s) missing, damaged or deformed	Visual

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	Defect	Method of Inspection
MAJOR		
101.	Stock height exceeds maximum allowable	Gage
102.	Insecure assembly	Manual
103.	Rotor drive gear assemblies not fully wound and cocked	Visual
104.	Escapement spring broken, deformed or unseated	Visual
105.	End of "D" shaft spring protrudes beyond diameter	Visual
106.	Parts missing or improperly assembled	Visual
107.	Escapement lever not locked on proper side of pallet lock slides	Visual
108.	Centrifugal gear springs project into body assembly screw holes	Visual
109.	Metal defective	Visual
110.	Pins or protrusions above flush of either surface	Visual
111.	Foreign matter	Visual
MINOR		
	None defined	

4.3.2.4 Movement Subassembly (after dynamic regulation except as other specified), Drawing F8593032, covering a detail of Drawing F8595229.

CRITICAL		
1.	Firing pin fires on "not fire" position of firing arm	Gage
2.	Extension of firing arm upright above top plate less than minimum allowable (with firing arm shaft shouldered on bottom of plate No. 2)	Gage
3.	Lever assembly not engaged in arbor collar slot	Manual
MAJOR		
101.	Firing pin does not fire on "must fire" position	Gage
102.	Firing pin protrusion less than min., in fired position	Gage
103.	Firing arm shaft spring protrudes	Visual
104.	Insecure assembly	Manual
105.	Parts missing or improperly assembled	Visual
106.	Foreign matter	Visual
107.	Staking of movement assembly screws inadequate (prior to regulation)	Visual

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Defect	Method of Inspection
MINOR	
201. Firing arm does not advance to "minimum travel" position	Gage
202. Staking indentations for timing disc retainer inadequate (prior to regulation)	Visual

4.3.2.5 Timing Disc Assembly, Drawing C8594472, covering a detail of Drawing F8595229.

CRITICAL	
1. Safety disc tabs not engaged in timing disc bushing slots	Visual
MAJOR	
101. Adjusting segment radius extends beyond periphery of timing disc	Visual
102. Safety disc radius below periphery of timing disc	Visual
MINOR	
None defined	

4.3.2.6 Movement Assembly, Drawing F8594465 covering a detail of Drawing F8595229.

CRITICAL	
1. Timing disc bushing loose on knurl of arbor (check prior to assembly of locknut)	Manual
2. Timing disc retainer insecure, damaged or missing	Manual & Visual
3. Distance from top of top plate to top of retainer exceeds maximum allowable (after staking of retainer)	Gage
MAJOR	
101. Total height exceeds max. allowable	Gage
102. Clearance between plate No. 7 and safety disc less than minimum allowable	Gage
103. Clearance between adjusting segment and underside of retainer less than minimum allowable	Gage
104. Escapement lever not locked on proper side of pallet locks	Visual
105. Escapement spring broken, deformed or unseated	Visual
106. Pins or protrusions above flush of bottom surface	Visual

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Defect	Method of Inspection
MINOR	
None defined	

4.3.2.7 Timers, numbers 1, 2, and 3 subassemblies, Drawings D8594470, D8594471, and D8594463 covering details of Drawing F8595229.

CRITICAL		
1.	Edge of firing arm outside of safety disc setting lines	Visual
MAJOR		
101.	Eccentricity/nonperpendicularity of arbor shaft	Gage
102.	Nonparallelism of timing disc to surface "B"	Measure
103.	Distance from firing arm upright to surface "B" less than minimum allowable	Gage
104.	Height of arbor above surface "B" exceeds max.	Gage
105.	Insecure assembly	Manual
106.	Metal defective	Visual
107.	Foreign matter	Visual
108.	Burr	Visual
MINOR		
None defined		

4.3.2.8 Dial Assemblies, Drawings D8594467, D8594468, and D8594469, covering a detail of Drawing F8595229.

CRITICAL		
None defined		
MAJOR		
101.	Setting pin extension beyond surface A exceeds max. allowable	Gage
102.	Insecure assembly	Manual
103.	Hammerspring rubs against setting pin	Visual
104.	Part(s) missing or improperly assembled	Visual
105.	Metal defective	Visual
MINOR		
None defined		

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4.3.2.9 Timer No. 1, 2, and 3 Assemblies, Drawings F8594396, F8594397, and F8594461 (before assembly of dial plug), covering details of Drawing F8595229.

	Defect	Method of Inspection
CRITICAL		
1.	Setting pin engagement with timing disc leaf less than minimum allowable with end play removed - arbor down, dial assembly up (check after lockwire tensioning operation)	Measure
2.	Staking of setting pin inadequate	Visual
3.	Dial assembly not seated properly in housing	Visual
MAJOR		
101.	Pellet holder location with respect to bottom surface of Timer No. 3 housing incorrect (Drawing F8594461)	Gage
102.	Clearance between adjusting stud and hammer-spring weights less than minimum allowable	Gage
103.	Insecure assembly	Manual
104.	Setting pin not engaged in timing disc leaf slot	Visual
105.	Incorrect zero setting	Visual
106.	Hammerspring not over lug when timer is set on Safe(s)	Visual
107.	Part(s) missing or improperly assembled	Visual
MINOR		
	None defined	

4.3.2.10 Fuze Assembly, Drawing F8594464 for the MT, M542 Fuze, covering a detail of Drawing F8595229.

	Defect	Method of Inspection
CRITICAL		
1.	Timer assemblies not set at safe(s) position	Visual
MAJOR		
101.	Total length exceeds maximum allowable	Gage
102.	Major diameter of external thread exceeds max. allowable	Gage
103.	Pitch diameter of external thread incorrect	Gage
104.	Outside diameter of fuze body exceeds max. allowable	Gage
105.	Eccentricity of housing(s) with respect to external thread	Measure

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Defect	Method of Inspection
MAJOR, Cont.	
106. Insecure assembly	Manual
107. Gap between housings or at breech block interface	Visual
108. Marking missing, incorrect or illegible	Visual
109. Part(s) missing or incorrectly assembled	Visual
MINOR	
201. Minor diameter of internal thread less than minimum	Gage
202. Pitch diameter of internal thread incorrect	Gage

4.3.2.11 Fuze, MT, M542, with Igniter Assembly, Drawing F8595229.

CRITICAL	
None defined.	
MAJOR	
101. Overall length exceeds maximum allowable	Gage
102. Inadequate assembly; loose or missing parts	Manual/Visual
103. Staking (of igniter to fuze) inadequate	Visual
104. Metal defective	Visual
105. Foreign matter	Visual
106. Burr	Visual

4.3.2.12 Ammunition Container, Metal, Drawing F8794406.

CRITICAL	
None defined.	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Key missing or damaged	Visual
203. Container punctured, dented, damaged or deformed	Visual
204. Marking missing, incorrect or illegible	Visual

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Defect	Method of Inspection

4.3.2.13 Box, Packing, Drawing B8794405.

CRITICAL

None defined.

MAJOR

None defined.

MINOR

201.	Contents loose	Manual
202.	Hardware missing, broken, loose or improperly engaged	Manual/Visual
203.	Rope handles insecure, damaged, or missing	Manual/Visual
204.	Strapping missing, broken, loose or improperly assembled	Manual/Visual
205.	Car seal missing, unsealed or improperly positioned (when applicable)	Visual
206.	Box damaged	Visual
207.	Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (Major - Destructive).- Twenty samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

Component	Drawing Number
Gear, Escapement	C8594412
Arbor, Excapement	D8594298
Pinion	C8593087
Pin, Lever Spring	B8593144
Pin, Locating	B8593119
Pin, Anchor	B8593120
Mainspring	C10521340
Pin, D shaft	B8594439
Pin, Pallet Lock Slide	B8593017
Pin, Firing Arm Shaft	B8593022
Pinion No. 4	B8593113
Lever, Escapement (Movement)	F8593023
Arm, Firing	C8593095
Guide, Adjusting	B8594273
Lever, Escapement (Delay Arming Mechanism)	B85944400

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4.3.3.1, Cont.

Component	Drawing Number
Spring, Hammer	B8594431
Spring, Timing Disc	B8594454
Washer, Timing Disc	B8594267

4.3.3.2 Static load and torque tests (Major, Destructive).- The sample size and criteria for acceptance shall be as specified for range plans in MIL-STD-414. An AQL of 0.25% shall be used. If the variables analysis indicates rejection but no individual value has actually failed to meet the drawing requirements, a comparable attributes plan in MIL-STD-105 shall be used, in which case the sample units used for the variables test shall be considered a part of the attributes sample.

4.3.3.2.1 Static load tests (Major, Destructive).

Assembly	Drawing Number
Centrifugal Drive Gear Assembly	C8594473, -414
"D" Shaft Assembly	C8594415
No. 1 Gear and Pinion Assembly (D.A.M.)	C8594418
No. 2 Gear and Pinion Assembly (D.A.M.)	C8594419
No. 3 Gear and Pinion Assembly (D.A.M.)	C8594420
Escapement Gear & Pinion Assembly (D.A.M.)	C8594411
Escapement Lever Assembly (D.A.M.)	D8594406
Pallet Lock Assembly (D.A.M.)	D8593152
Rotor Assembly (D.A.M.)	D8594410
Firing Arm Assembly (Pin & Arm)	C8593094
Escapement Gear & Pinion Assy (Movement)	C8593382
Escapement Lever Assembly (Movement)	F8593015
Pallet Lock Assembly (Movement)	B8593016
Lever Assembly (Movement)	B8593124
No. 1 Gear & Pinion Assembly (Movement)	C8594280
No. 2 Gear & Pinion Assembly (Movement)	C8594283
No. 3 Gear & Pinion Assembly (Movement)	C8594286
No. 4 Gear & Pinion Assembly (Movement)	C8593085
No. 5 Gear & Pinion Assembly (Movement)	C8593090
Disc and Stud Assembly (Movement)	C10520614

4.3.3.2.2 Torque tests (Major, Destructive)

Assembly	Drawing Number
Escapement Gear & Pinion Assembly (D.A.M.)	C8594411
Escapement Lever Assembly (D.A.M.)	D8594406
Escapement Gear & Pinion Assembly (Movement)	C8593382
Escapement Lever Assembly (Movement)	F8593015
No. 5 Gear & Arbor Assembly (Movement)	C8593090

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4.3.3.3 Dead load tests (Major, Destructive).

4.3.3.3.1 Timing disc spring.- The same 20 springs that comply with the hardness test shall be subjected to the dead-load test. Failure of any spring to conform to the requirement specified on Drawing B8594454 shall be cause for rejection of the lot.

4.3.3.3.2 Firing pin spring.- Twenty springs from each lot shall be subjected to the compression test. Failure of any spring to comply with the requirement specified on Drawing B8594315 shall be cause for rejection of the lot.

4.3.3.4 Hammerspring deflection test (Critical, Nondestructive).- Each of the dial assemblies denoted below shall be tested for compliance with the spring deflection requirement of the applicable drawing. Any assembly failing to conform to the requirement shall be rejected.

Assembly	Drawing Number
Upper Dial Assembly	D8594467
Middle Dial Assembly	D8594468
Lower Dial Assembly	D8594469

4.3.3.5 Firing arm bend test (Major, Destructive).- The same 20 firing arms that comply with the hardness requirements shall be subjected to the bend test. Failure of any firing arm to comply with the requirement specified on Drawing C8593095 shall be cause for rejection of the lot.

4.3.3.6 Kinetic energy of firing pin (Major, Nondestructive).- Fifty movement subassemblies from each lot shall be tested to determine compliance with the requirement specified on Drawing F8593032. Failure of any assembly to conform to the requirement shall be cause for rejection of the lot.

4.3.3.7 Arming test.- Movement Subassembly (Major, Nondestructive).- The pallet lock and lever assemblies within each movement subassembly shall be tested to determine compliance with the arming requirement of Drawing F8593032. Any movement subassembly failing to conform to the requirement shall be rejected.

4.3.3.8 Nonarming test-movement subassembly (Critical, Nondestructive).- The pallet lock and lever assemblies within each movement subassembly shall be tested to determine compliance with the nonarming requirement of Drawing F8593032. Any movement subassembly failing to conform to the requirement shall be rejected.

4.3.3.9 Friction torque tests.

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4.3.3.9.1 Disc and stud torque (Minor, Nondestructive).- Twenty disc and stud assemblies shall be tested for compliance with the torque requirement of Drawing C10520614. Failure of any assembly to conform with the requirement shall be cause for rejection of the lot.

4.3.3.9.2 Timing disc torque (Critical, Nondestructive).- Each movement assembly shall be tested for compliance with the torque requirement of Drawing F8594465. Any movement assembly failing to conform to the requirement shall be rejected.

4.3.3.9.3 Setting torque.- Timer No. 1, 2, and 3 Assemblies (Critical Nondestructive).- Each assembly shall be tested for compliance with the torque requirement of Drawing F8594396, F8594397, and F8594461, respectively. Any assembly failing to conform to the requirement shall be rejected.

4.3.3.10 Arming.- Delay Arming Mechanism (Major, Nondestructive).- Each unarmed delay arming mechanism shall be tested to determine compliance with the arming requirement of Drawing F8594387. Any assembly which is not armed after spinning (see 6.2.1) shall be rejected.

4.3.3.11 Nonarming.- Delay Arming Mechanism Assembly (Critical, Nondestructive).- Each unarmed delay arming mechanism shall be tested to determine compliance with the nonarming requirement of Drawing F8594387. Any assembly failing to remain unarmed after spinning (see 6.2.2) shall be rejected.

4.3.3.12 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

4.3.3.12.1 Bending adjustment of firing arm upright is not permitted at any time.

a. Packing Box, Drawing F8794405, and Metal Container, Drawing F8794406.

   (1) Packing operations (Minor)

b. Fuze, Mechanical Time, M542 with Igniter, Assembly Drawing F8595229.

   (1) Application of protective coating (Minor)

   (2) Correct assembly of cones and disc (Major)

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c. Timer No. 1, 2, and 3 Assemblies, Drawings F8594396, F8594397, and F8594461.

(1) Setting pin engagement (Critical)

(2) Zero setting procedure (Major)

(3) Insertion of pellet screw assembly in Timer No. 3 (Major)

(4) Application of sealant (Minor)

d. Delay Arming Mechanism, Drawing F8594387

(1) Application of lubricant (Major)

(2) Removal of gear-train backlash (Major)

(3) Assembly screw torquing operation (Major)

(4) Winding and cocking operation (Major)

e. Plate No. 1 and 2 Assembly, Drawing D8594404

(1) Application of lubricant (Major)

f. Movement Assembly, Drawing F8594465

(1) Locknut torquing operation prior to timing disc torque test (Critical)

g. Movement Subassembly Drawing F8593032

(1) Staking of top assembly screw and torquing of bottom assembly screws prior to dynamic regulation (Major).

(2) Application of lubricant (Major)

(3) Winding of mainspring before assembly of screw (Major)

(4) Dynamic regulation operationg (Major)

(5) Winding of movement after dynamic regulation (Major)

(6) Gear-train backlash removal (Major)

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h. Laminated Plates No. 1 and No. 2 Assembly, Drawing F8593043

(1) Application of lubricant (Major)

4.3.3.13 Environmental tests (Critical, Destructive).- Fifteen (15) fuze assemblies (Drawing F8595229), shall be selected from the initial production sample and subjected to the following tests. Each fuze shall be set on Safe(S). Failure of any fuze to comply with the requirements with regard to safety shall be cause for rejection. Evaluation of the fuzes subjected to transportation-vibration shall include nonarming tests of the delay arming mechanism and movement subassembly per 4.3.3.11 and 4.3.3.8, respectively.

Environmental Test	Quantity to Test	Requirement Paragraph
Jolt	3	3.3.1
Jumble	4	3.3.2
Forty-foot drop	5	3.3.3
Transportation-Vibration	3	3.3.4

4.3.3.13.1 Environmental-ballistic tests (Critical, Destructive).- The Government will subject 9 fuzes, represented by 27 test fuzes (see 6.3.4) to the environmental-ballistic tests specified in 4.4.18. The occurrence of a premature burst, or more than two time failures, shall be cause for rejection of the initial production sample.

4.3.3.14 Ballistic tests (Critical, Destructive).- The Government will subject 21 fuzes, represented by 63 test fuzes (see 6.3.4.) to the ballistic tests of 4.4.19. The occurrence of any of the following shall be cause for rejection of the lot which the ballistic sample represents.

a. A premature burst (see 6.3.1)

b. An off-time fast (see 6.3.2)

c. More than two (2) off-times slow (see 6.3.2)

d. More than one (1) time failure (see 6.3.3)

e. Failure of the sample statistics to comply with the requirements of Table I.

4.3.3.16 Air pressure, container (Major, Nondestructive).- Each sealed metal container shall be subjected to an air pressure test. Any container which fails to comply with the requirements of MIL-C-10464 shall be rejected.

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4.3.4 Inspection equipment.- The examinations and tests shall be made using equipment listed in IEL-8595229. Equipment design shall be in accordance with the code designation on the applicable IEL (reference is made to 6.5 for definition of the coding format).

4.4 Test methods and procedures.

4.4.1 Hardness.- Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three (3) hardness readings taken on each sample unit shall be used as the individual value.

4.4.2 Destructive static load.- Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. All units shall be tested to the point of component disassembly. The values obtained shall be used to determine lot acceptability in accordance with the specified variables plan. All samples tested shall be scrapped.

4.4.3 Destructive torque.- Escapement lever assemblies shall be tested prior to insertion of the escapement spring. All units shall be tested to the breakaway point. The values obtained shall be used to determine lot acceptability in accordance with the specified variables plan. All samples tested shall be scrapped.

4.4.4 Timing disc spring dead-load.- The test shall be performed using the values as specified on Drawing B8594454. The timing disc springs that have been subjected to the dead-load test shall be scrapped.

4.4.5 Firing pin spring compression.- The firing pin springs shall be compressed for the length of time specified on Drawing B8594315. The springs that have been subjected to the compression test shall be scrapped.

4.4.6 Hammerspring deflection.- Upper, Middle and Lower Dial Assemblies.- The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on Drawings D8594467, D8594468 and D8594469. The hammerspring must return to surface "A" when released.

4.4.7 Firing arm bend.- The test shall be performed to determine compliance with the requirement on Drawing C8593095. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.8 Kinetic energy of firing pin (Movement Subassembly).- The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position. At the conclusion of the test, each subassembly shall be recocked.

4.4.9 Arming (Movement Subassembly).- The arming test shall be performed on completely wound subassemblies. The subassemblies shall be rotated at the r.p.m. specified on Drawing F8593032 in a clockwise direction when viewed from the bottom, for a period of not less than 15 seconds at constant r.p.m. to determine compliance with the requirement. At the conclusion of the test, each mechanism shall be rewound. The arming test shall be performed prior to the nonarming test.

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4.4.10 Nonarming (Movement Subassembly).- The nonarming test shall be performed on completely wound subassemblies. The subassemblies shall be rotated at the r.p.m. specified on Drawing F8593032 in a clockwise direction when viewed from the bottom, for a period not less than 15 seconds at constant r.p.m. to determine compliance with the requirement. This test shall follow the arming test and shall be the last spin test to which the movement is subjected prior to assembly into the fuze.

4.4.11 Disc and stud torque.- The disc and stud assemblies shall be tested for function torque to determine compliance with the requirement of Drawing C10520614.

4.4.12 Timing disc torque.- The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest readings will be taken in one complete revolution of the timing disc for determination of compliance with the requirements of Drawing F8595465.

4.4.13 Setting torque (Timer No. 1, 2, and 3 Assemblies).- Each timer assembly shall be tested for setting torque between the dial and housing to determine compliance with the applicable drawing requirements.

4.4.14 Arming (Delay Arming Mechanism).- Each unarmed mechanism shall be placed in a spin fixture and rotated at the spin speeds specified on Drawing F8594387. The time required to arm the assembly shall be recorded for each spin speed specified to within $\pm$ 0.1 second. Prior to and at the conclusion of each speed run, the assembly shall be fully wound and reset. This test shall precede the nonarming test.

4.4.15 Nonarming (Delay arming mechanism assembly).- The nonarming test shall be performed on completely wound (unarmed) delay arming mechanism assemblies. The unarmed mechanism shall be rotated at the RPM specified to determine conformance with the requirement of Drawing F8594387. This test shall follow the arming test and shall be the last spin test to which the delay arming mechanism is subjected prior to assembly into the timer.

4.4.16 Dynamic regulation monitoring.- The following procedure shall be observed: The movement subassemblies shall be regulated, using the operational procedures furnished with the equipment. The movement subassembly shall be rotated in a clockwise direction when viewed from the bottom at a constant speed of 6,000 $\pm$ 25 r.p.m. Timekeeping and performance shall be checked over a period of not less than 100 seconds. No adjustment shall be made on the escapement during the 100 second check run. Inspection shall be performed on a monitoring basis only (4.3.3.13).

4.4.17 Environmental test procedures.

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4.4.17.1 Jolt.- The fuze shall be jolted in accordance with MIL-STD-350, except that the fuze shall not be jolted in the horizontal position. After disassembly and examination, all components shall be scrapped.

4.4.17.2 Jumble.- The fuze shall be jumbled in accordance with MIL-STD-351, except that the fuze shall be jumbled in a metal packing container. After disassembly and examination, all components shall be scrapped.

4.4.17.3 Forty (40) foot drop.- The fuze shall be dropped in accordance with MIL-STD-352, except that the fuze shall be dropped in a metal packing container. After disassembly and examination, all components shall be scrapped.

4.4.17.4 Transportation-vibration.- The fuze shall be tested in accordance with MIL-STD-353. All disassembled components shall be scrapped.

4.4.18 Environmental-ballistic tests.- Three (3) groups of three (3) fuzes each, including three (3) fuzes which have been subjected to the transportation-vibration environment of MIL-STD-353, shall be reassembled as 27 test fuzes (see 6.3.4) and proportionately allocated for ballistic firing under the following conditions, with the fuze, projectile and propellant maintained at the temperature indicated.

Environmental Ballistic Test	Test Fuze Quantity	Test Fuze Setting	Weapon	Charge	Temperature
Post Vibration-Ambient Temp.	9	55 Sec.	8" How.	Zone 7	$70^{\circ} \pm 10^{\circ}$ F
Low Temparature	9	20 Sec.	8" How.	Zone 1	$-40^{\circ} \pm 5^{\circ}$ F
High Temperature	9	55 Sec.	8" How.	Zone 7	$160^{\circ} \pm 5^{\circ}$ F

Time-to-burst shall be recorded for information purposes, using two independent timing devices capable of recording to the nearest millisecond.

4.4.19 Ballistic tests.- Twenty-one (21) fuzes, disassembled to the extent of removing all timer assemblies intact, shall be reassembled to provide 63 test fuzes (see 6.3.4) and allocated for ballistic firing under the following conditions, with the temperature of the fuze, projectile and propellant maintained at $70^{\circ} \pm 10^{\circ}$ F.

Fuze Quantity	Setting	Weapon	Charge
10 fuzes (represented by 30 test fuzes)	20 Sec	8" How.	Zone 1
11 fuzes (represented by 33 test fuzes)	55 Sec	8" How.	Zone 7

Only those values which are beyond $\pm$ 3 sample standard deviations from the sample mean value will be excluded in the final computation of the mean and standard deviation statistics. Time-to-burst shall be recorded to the nearest millisecond, using two independent timing devices each capable of recording to the nearest millisecond.

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4.4.20 Air pressure test.- Each sealed metal container shall be tested for air leakage according to MIL-C-10464.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Packaging and packing shall be in accordance with Drawings F8794406 and B8794405, unless otherwise directed by the contracting officer.

5.2 Marking.

5.2.1 Metal container.- The metal container shall be marked in accordance with the specific instructions denoted on Drawing F8794406.

5.2.2 Packing box.- The box shall have the I.C.C. nomenclature, "Time Fuzes - Handle Carefully," prominently stenciled on the cover. The following information shall be stenciled on the cover or placed on a label securely fastened to the cover.

a. Quantity and nomenclature of item

b. Part number

c. Lot number and date loaded

d. Contract number and name of manufacturer

e. Name and address of consignor

f. Name and address of consignee

The quantity and nomenclature of the item, part number, and lot number shall also be placed on one end of the box.

5.3 Data cards.- Data card information shall be as specified in MIL-A-2550.

6. DEFINITIONS AND NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. It is recognized that manufacturing capabilities using current mass production methods will not fully provide consistent compliance with all dimensional requirements. Dimensions which are specifically required by Classification of Defects as listed in Section 4 or those which affect external profile of assembly are mandatory, however, the contractor at his own risk may utilize components exceeding the specified dimensional limits not included in the

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foregoing, provided there is no adverse effect on fuzes functionability. This relaxation of requirements applies to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.1 The contractor shall maintain records showing relaxations from dimensional requirements, the extent of such relaxations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Armed/unarmed.

6.2.1 Armed.- Locked (spring-loaded) in a position in which the detonator is entirely exposed in the hole into which the firing pin enters and the flash emanates.

6.2.2 Unarmed.- Locked (spring-loaded) is a position in which the detonator is not at all exposed in the hole into which the firing pin enters and the flash emanates.

6.3 Ballistic parameters.

6.3.1 Premature burst.- Shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun.

b. The round detonates before 75% of set time.

c. The round detonates before 8.6 seconds has expired.

6.3.2. An off time, either fast or slow, is an individual air burst which occurs outside the upper or lower limits, respectively, which are specified for individual times of flights (Table I).

6.3.3 A time failure is considered to have occurred if the round fails to detonate during flight

6.3.4 Test fuze.- A test fuze consists of one timer assembly arranged with two dummy complements, with the original orientation of the timer maintained (see Drawing F10523262), and contained for firing in the test shell, (Drawing D10523268.)

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6.4 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, and packing.

(1) Level "B" preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined the fuzes will be held in covered storage approximately two months from date of initial packaging.

6.5 Inspection equipment lists (IEL) codes.-

(AD) - Army design.- Army designs are detailed drawings which depict completely all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an army design is listed for an item of inspection equipment.

(SCD) - Specification control drawing.- Specification control drawing depicts the minimum equipment requirements in outline. Descriptive, diagrammatic or pictorial form specifies required performance or other characteristics. Such drawings are provided by the Government as an attachment to the pertinent inspection equipment lists. Approval of items of inspection equipment fabricated in accordance with these drawings is delegated to the Inspection Element of the Army Procurement District administering the contract.

(CE) - Commercial equipment.- Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the Inspection Element of the Army Procurement District administering the contract.

(CD) - Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate, and operate an item of inspection equipment. This requires that the necessary view, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Such designs shall be submitted for approval to the Commanding Officer, Frankford Arsenal, ATTN: SMUFA-4920, Philadelphia, Pa., 19137.

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Army - MU

Preparing activity:

Army - MU

Project No. 1390-A280

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